Lecture 4: Calculus for Data Science

Math for Data Science · 2026

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2026-09-22

Agenda

- Part I: Functions and the Derivative
- Part II: Differentiation
- Part III: L'Hôpital's Rule
- Part IV: Derivatives and the Shape of a Graph
- Part V: Integration

Part I: Functions and the Derivative

Introducing Calculus

It is impossible to teach differential calculus in a single session, so here is what we will do:

- Take a bird's-eye view of calculus and how we will use it
- Provide a map of the territory that you should (eventually) be comfortable with
- Work through some applied differentiation skills

Calculus: A Bird's Eye View

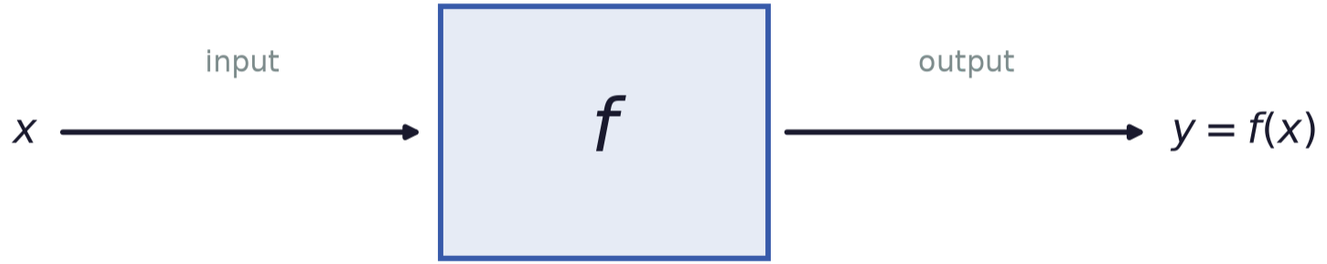
Before we even talk about calculus, let's remind ourselves about functions:

- A function is a **map** from some input x to some output y.

Some notable functions include:

- A probability mass function (PMF) and its continuous analog, the probability density function (PDF): these map events to the probabilities of those events occurring
- A regression: a function describing how predictors $x_1, x_2, x_3, \ldots$ map to some outcome y
- A loss function: in predictive modeling we try to minimize some kind of prediction error, captured by a function

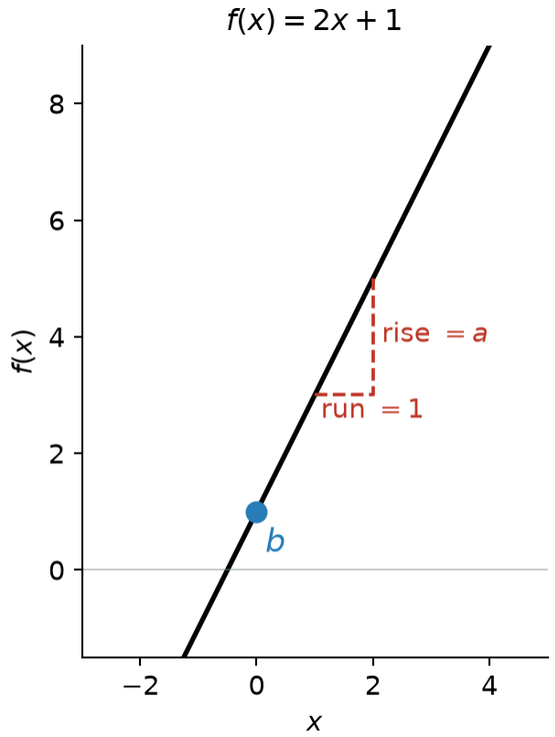
A Function Is an Input-Output Device



A function takes an input, does something to it, and returns exactly one output.

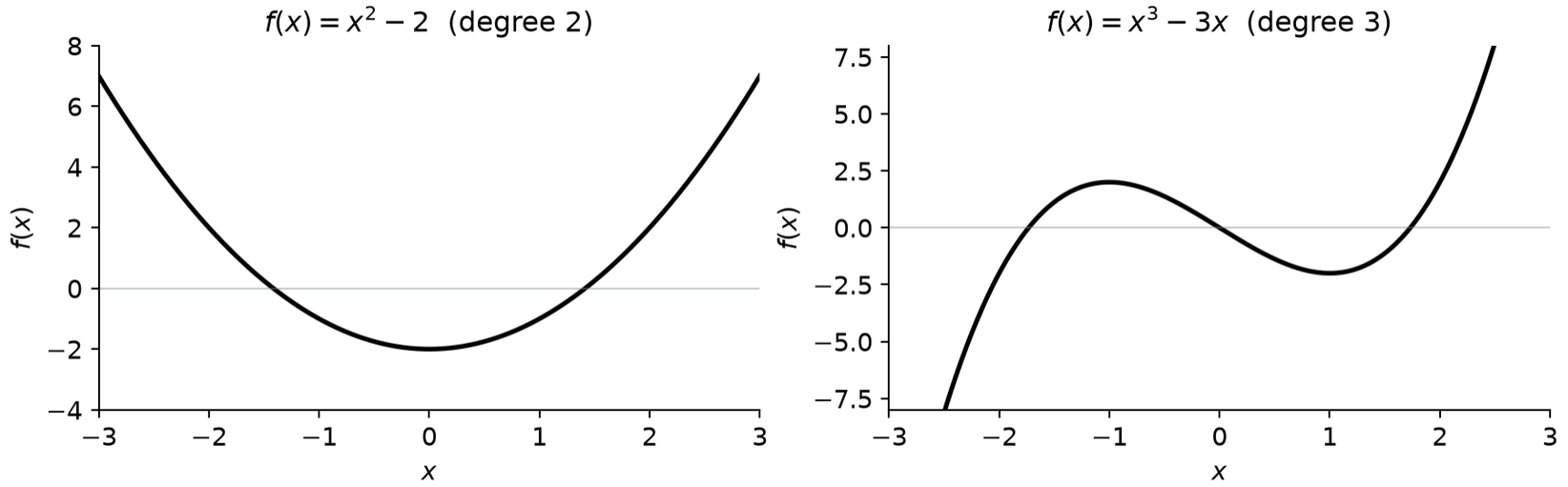
Some Functions You Should Know

A linear function takes the form $f(x) = ax + b$, with a the slope and b the y-intercept.



Some Functions You Should Know

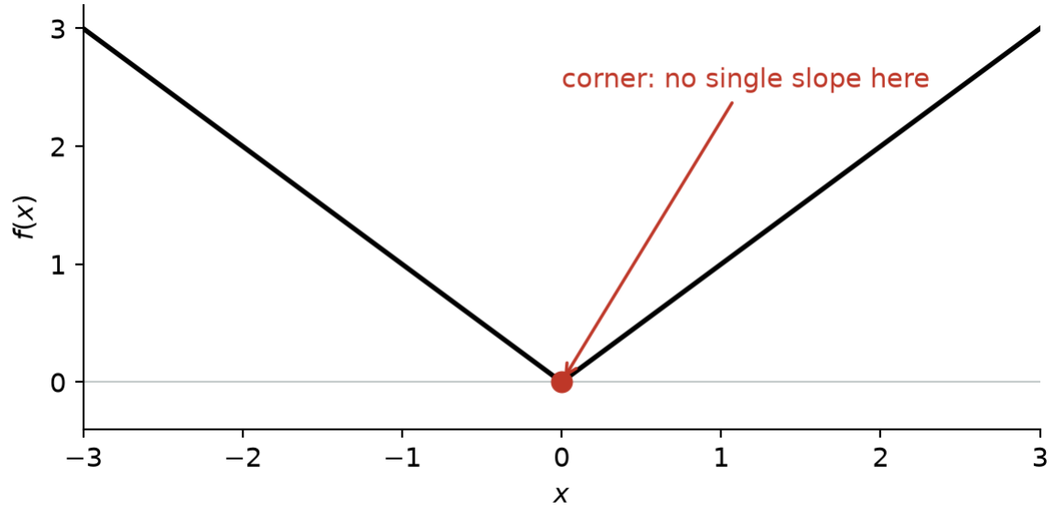
A polynomial function takes the form $f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$.



The degree n is the highest power - it caps how many times the curve can turn.

Some Functions You Should Know

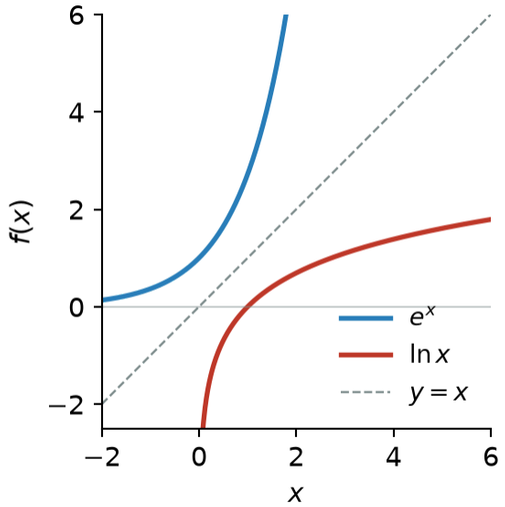
The absolute value function takes the form $f(x) = |x|$.



Keep that corner in mind - it is the standard example of a function that is continuous but *not* differentiable everywhere.

Some Functions You Should Know

The exponential function takes the form $f(x) = a^x$, and the logarithmic function "undoes" it: $f(x) = \log_b(x) = y$ if and only if $b^y = x$.



They are reflections of each other across the line $y = x$.

What Calculus Adds

Calculus deals with the rates of change of functions.

- And *differential* calculus finds the **instantaneous** rate of change at a point - which is itself a function.

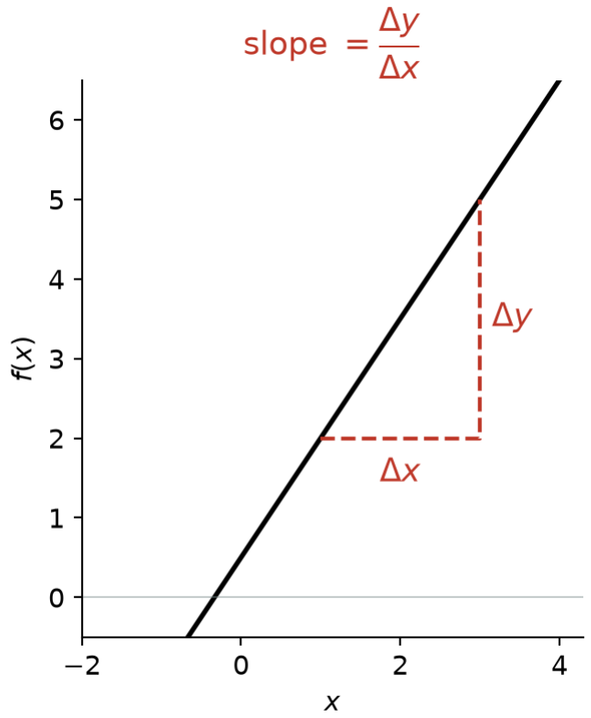
When is this useful?

- A PDF is the derivative of a CDF - exactly what we need for continuous random variables
- In predictive modeling we often minimize a loss function, which captures the distance from our model's predictions to the observed data
  - The minimum of a function is achieved where the first derivative is 0

Part II: Differentiation

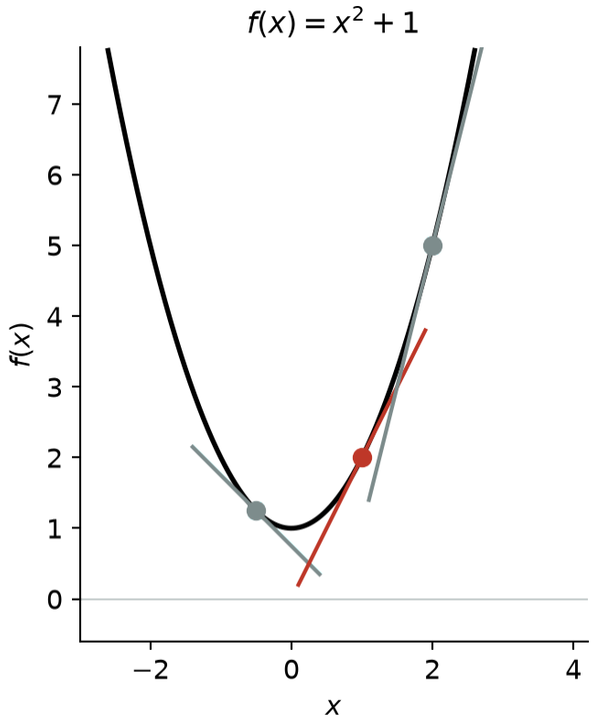
If Your Function Is a Line, It’s the Slope

So how do we find the instantaneous rate of change of a function at a particular point?



For a line the rate of change never varies, so one number describes the whole function.

And If Your Function Is a Parabola?



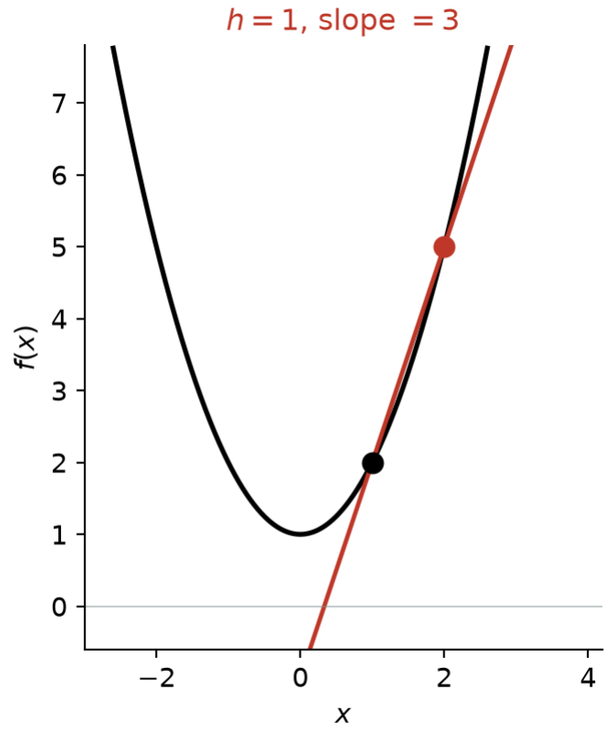
The rate of change is now different at every point: it is the slope of the tangent line there.

How Do We Find the Slope of the Tangent Line?

Take an arbitrary second point on the curve and run a line through both. Its slope is

$$\frac{f(x+h) - f(x)}{h}$$

Now shrink h:

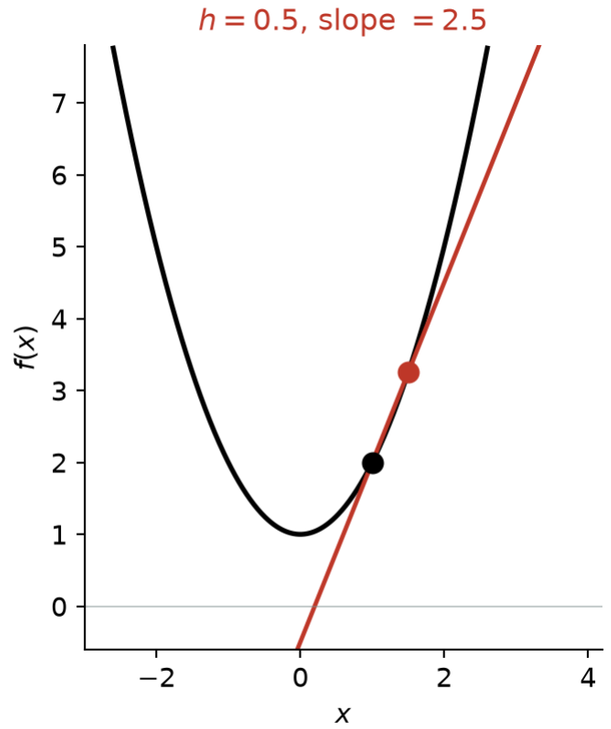


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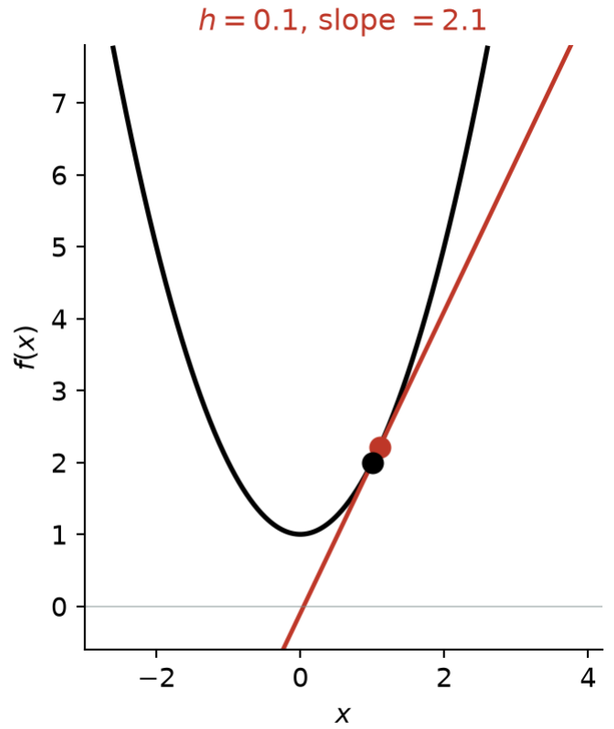


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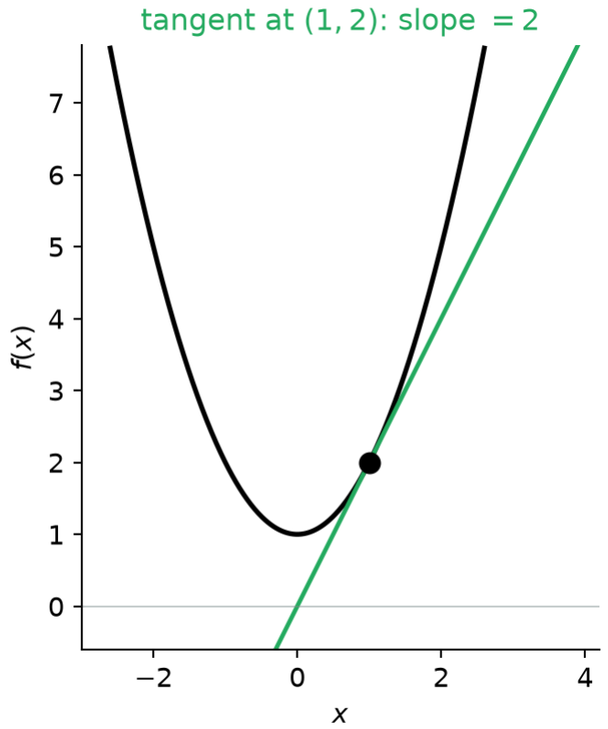


How Do We Find the Slope of the Tangent Line?

Take an arbitrary second point on the curve and run a line through both. Its slope is

$$\frac{f(x+h)-f(x)}{h}$$

Now shrink h:



The Secant Slopes Converge

h	Slope of the line from $(1, 2)$ to $(1+h,\ f(1+h))$
1	3
0.5	2.5
0.1	2.1
0.01	2.01
0.001	2.001

As $h \longrightarrow 0$, the slope converges to 2. And indeed

$$\frac{f(1+h)-f(1)}{h} = \frac{(1+h)^2+1-2}{h} = \frac{2h+h^2}{h} = 2+h$$

Defining the Derivative

As $h \to 0$, the line through x and $x + h$ *becomes* the line tangent to the curve at x. So we define the derivative of $f(x)$ as

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

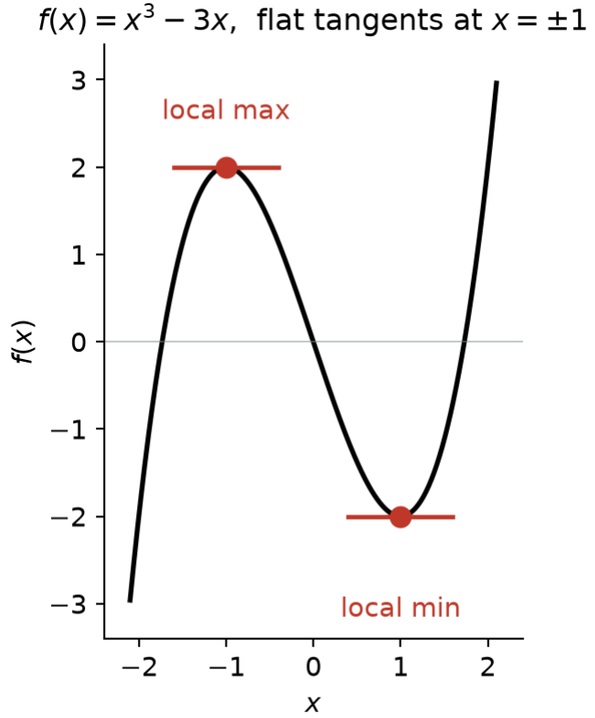
Here $f'(x)$ is the function giving the instantaneous rate of change of f at x.

Notation you will meet, all meaning the same thing:

$$f'(x) \qquad \frac{df}{dx} \qquad \frac{d}{dx} f(x) \qquad \dot{f}$$

Bookmark This for Later

At a maximum (and at a minimum), a function's derivative is 0.

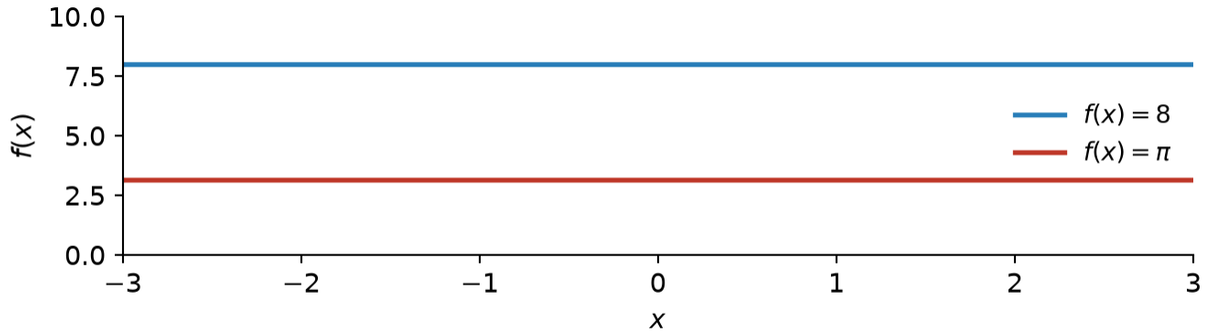


We will use this many, many times - deriving regression coefficients, maximizing likelihoods, minimizing loss functions.

Rule 1: The Constant Rule

$$\frac{d}{dx}c = 0$$

- What is the derivative of $f(x) = 8$? 0.
- The derivative of $f(x) = \pi$? 0.



A constant function never changes, so its rate of change is zero everywhere.

Rule 2: The Power Rule

$$\frac{d}{dx} x^n = n x^{n-1}$$

- What is the derivative of $f(x) = x^3$? $3x^2$.
- The derivative of $f(x) = 2x^2$? $4x$.
- The derivative of $f(x) = x$? 1.

This works for any real n, not just positive integers:

$$\frac{d}{dx}\sqrt{x} = \frac{d}{dx} x^{1/2} = \tfrac{1}{2} x^{-1/2} \qquad \frac{d}{dx}\frac{1}{x} = \frac{d}{dx} x^{-1} = -x^{-2}$$

Rule 3: The Sum and Difference Rules

$$\frac{d}{dx}\Big(f(x)+g(x)\Big) = \frac{d}{dx}f(x) + \frac{d}{dx}g(x)$$

$$\frac{d}{dx}\Big(f(x)-g(x)\Big) = \frac{d}{dx}f(x) - \frac{d}{dx}g(x)$$

What is the derivative of $f(x) = 8x - 2x^3$?

$$f'(x) = 8 - 6x^2$$

Rule 4: The Constant Multiple and Product Rules

$$\frac{d}{dx}kf(x) = k\frac{d}{dx}f(x)$$

$$\frac{d}{dx}\Big(g(x)f(x)\Big) = g'(x)\cdot f(x) + g(x)\cdot f'(x)$$

What is the derivative of $(3x^2)(x^3 - 6x)$?

- By the product rule: $(6x)(x^3 - 6x) + (3x^2)(3x^2 - 6) = 6x^4 - 36x^2 + 9x^4 - 18x^2 = 15x^4 - 54x^2$
- We could also have multiplied it out first and used the sum rule: $\frac{d}{dx}(3x^5 - 18x^3) = 15x^4 - 54x^2$ ✓

Rule 5: The Quotient Rule

$$\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{g(x) \cdot f'(x) - f(x) \cdot g'(x)}{\left(g(x)\right)^2}$$

What is the derivative of $\frac{5x^2}{4x+3}$?

$$\frac{(4x+3)(10x) - (5x^2)(4)}{(4x+3)^2} = \frac{20x^2 + 30x}{(4x+3)^2}$$

Rule 6: The Chain Rule

If y is a function of u, and u is a function of x, then

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

What is the derivative of $(2x^3 + 2x - 1)^4$?

1. Identify the inner and outer functions: $u(x) = 2x^3 + 2x - 1, \; y(u) = u^4$
2. Find $\frac{dy}{du}$: $4u^3 = 4(2x^3 + 2x - 1)^3$
3. Find $\frac{du}{dx}$: $6x^2 + 2$
4. Put it together: $\frac{dy}{dx} = 4(2x^3 + 2x - 1)^3(6x^2 + 2)$

Rules 7 and 8: The Exponential and the Log

Rule 7 - the natural exponential function. If $y(x) = e^x$, then

$$\frac{dy}{dx} = e^x$$

e^x is its own derivative: the one function whose rate of change equals its height.

Rule 8 - the natural logarithm. If $x > 0$ and $y = \ln(x)$, then

$$\frac{dy}{dx} = \frac{1}{x}$$

Together with the chain rule these two carry most of the weight in maximum likelihood estimation, where we differentiate $\log$-likelihoods of exponential-family densities.

When a Function Has Two Inputs

Almost nothing we model depends on one number. A loss depends on every weight; a joint density $f_{X,Y}(x, y)$ has two arguments.

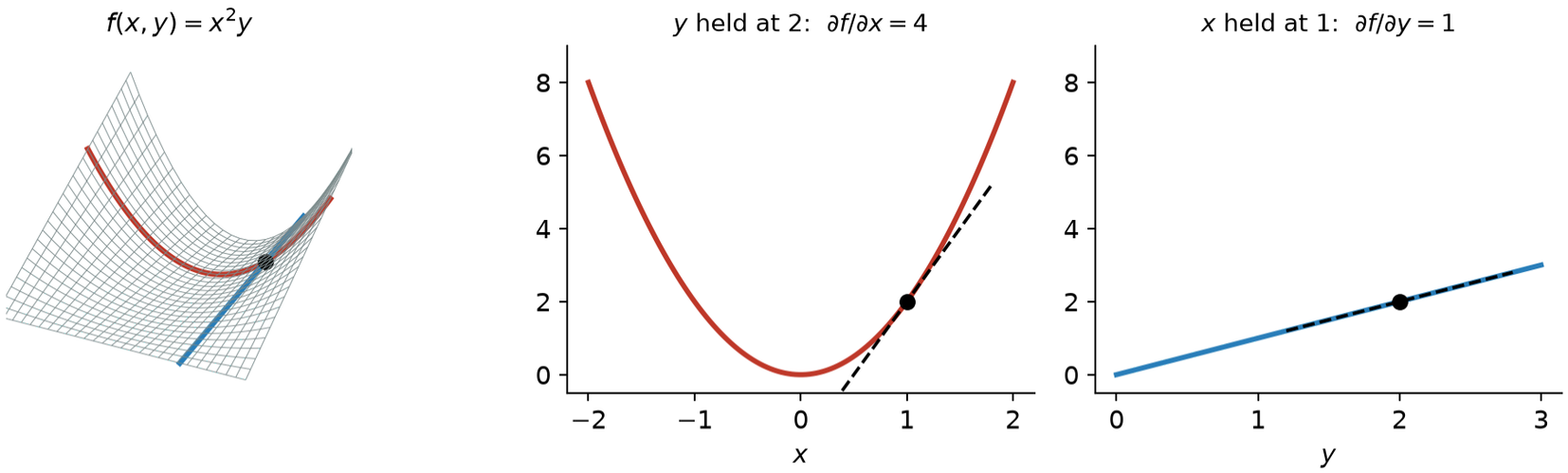
For $f(x, y)$ there is no single "the derivative" - there is one slope per direction you can move. Take them one at a time:

> The partial derivative $\frac{\partial f}{\partial x}$ is what you get by treating y as a **constant** and differentiating in x exactly as before.

Every rule from this part still applies. Nothing new has to be learned - only which letter is the variable this time.

Notation: Sometimes you will encouter $\partial_x f$ as a shorthand for $\frac{\partial f}{\partial x}$

What a Partial Derivative Reads Off a Surface



Holding y fixed cuts the surface along one line, leaving a curve in x alone - and $\partial f/\partial x$ is that curve's ordinary slope. Holding x fixed does the same in the other direction.

Partial Derivatives: Worked Examples

For $f(x, y) = x^2 y$:

$$\frac{\partial f}{\partial x} = 2xy \qquad\qquad \frac{\partial f}{\partial y} = x^2$$

In the first, y rides along as a constant multiple; in the second, x^2 is the constant and f is *linear* in y.

The rules carry over unchanged - here the chain rule, with y held fixed:

$$\frac{\partial}{\partial x} e^{xy} = y\, e^{xy} \qquad\qquad \frac{\partial}{\partial y} e^{xy} = x\, e^{xy}$$

Second Partials, and Why Order Does Not Matter

Differentiate twice, and you can do it in either variable. The mixed partial takes one of each:

$$\partial_x \partial_y f = \frac{\partial^2 f}{\partial x \, \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right)$$

For $f(x, y) = x^2 y$, taken both ways round:

$$\frac{\partial}{\partial x}\left(x^2\right) = 2x \qquad\qquad \frac{\partial}{\partial y}\left(2xy\right) = 2x$$

The same answer, and that is no accident: for any f whose second partials are continuous, **the order of differentiation does not matter**.

Part III: L’Hôpital’s Rule

A Limit the Rules Can’t Reach

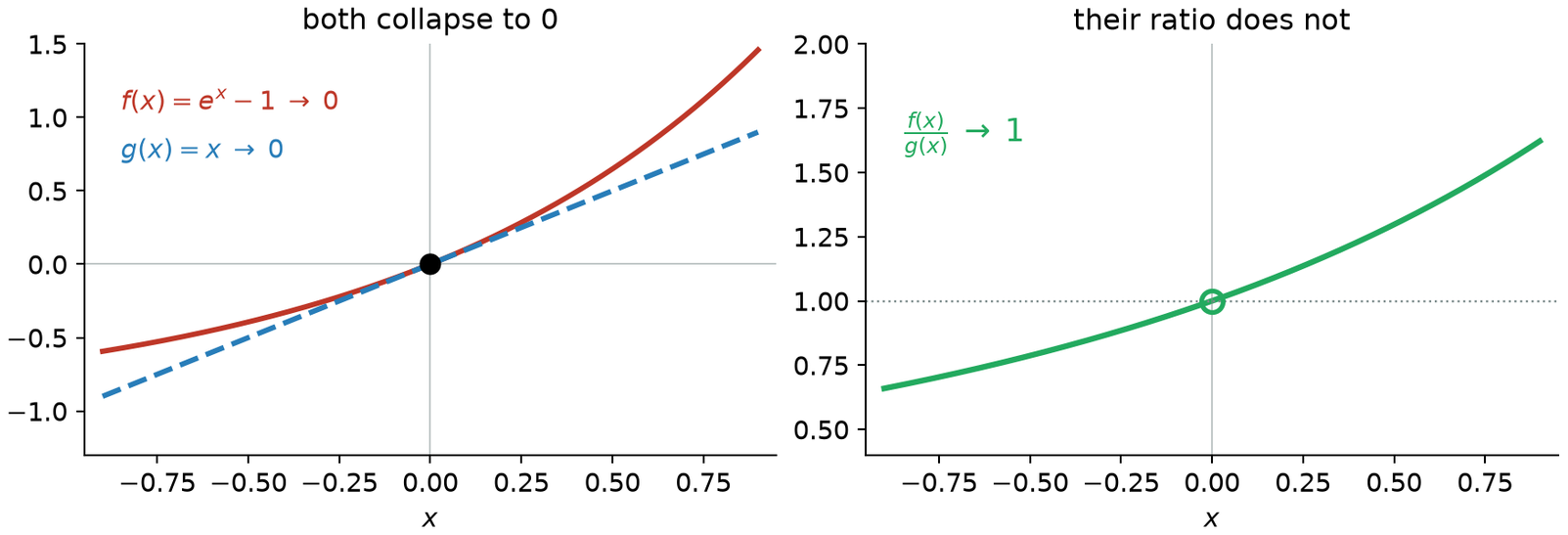
$$\lim_{x \to 0} \frac{e^x - 1}{x} = ?$$

Substituting $x = 0$ gives $\frac{0}{0}$ - not a number, and not an answer.

And none of the eight rules helps: the quotient rule *differentiates* a quotient, it does not *evaluate* one.

A Limit the Rules Can’t Reach

Yet the ratio itself is perfectly well behaved:



The hole at $x = 0$ is the only thing missing - and the limit is what fills it in.

Indeterminate Forms

$\frac{0}{0}$ is called an indeterminate form: the form alone does not decide the limit.

Limit	Form	Value
$\lim_{x\to 0} \frac{x}{x}$	$\frac{0}{0}$	1
$\lim_{x\to 0} \frac{5x}{x}$	$\frac{0}{0}$	5
$\lim_{x\to 0} \frac{x}{x^2}$	$\frac{0}{0}$	∞

Same form, three different answers. What matters is not *that* both parts vanish, but **how fast** each one does - and “how fast” is exactly what a derivative measures.

L'Hôpital's Rule

Let f and g be differentiable near a, with $g'(x) \neq 0$ there. If

$$\lim_{x\to a} f(x) = \lim_{x\to a} g(x) = 0 \qquad \text{or} \qquad \lim_{x\to a} f(x) = \lim_{x\to a} g(x) = \pm\infty$$

then

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}$$

whenever the limit on the right exists.

It holds just as well for $x \to \pm\infty$ and for one-sided limits.

Why It Works

Take the $\frac{0}{0}$ case, so $f(a) = g(a) = 0$. Near a, each function is close to its tangent line - the secant-to-tangent picture from Part II:

$$f(x) \approx f(a) + f'(a)(x-a) = f'(a)(x-a) \qquad g(x) \approx g'(a)(x-a)$$

Divide, and the $(x-a)$ that made both sides vanish cancels:

$$\frac{f(x)}{g(x)} \approx \frac{f'(a)(x-a)}{g'(a)(x-a)} = \frac{f'(a)}{g'(a)}$$

So the limit is a race between two slopes - which is why differentiating top and bottom *separately* is the right move here, and why it would be nonsense anywhere else.

Worked Examples

1. The opening limit. $\frac{0}{0}$, so differentiate top and bottom:

$$\lim_{x\to 0}\frac{e^x-1}{x} \overset{\text{L'H}}{=} \lim_{x\to 0}\frac{e^x}{1} = 1$$

2. $\lim_{x\to 1}\frac{\ln x}{x-1}$ is $\frac{0}{0}$:

$$\overset{\text{L'H}}{=} \lim_{x\to 1}\frac{1/x}{1} = 1$$

3. $\lim_{x\to\infty}\frac{x^2}{e^x}$ is $\frac{\infty}{\infty}$ - and stays indeterminate after one step, so apply the rule twice:

$$\overset{\text{L'H}}{=} \lim_{x\to\infty}\frac{2x}{e^x} \overset{\text{L'H}}{=} \lim_{x\to\infty}\frac{2}{e^x} = 0$$

Example 3 is worth remembering: the exponential outruns **any** polynomial.

Check the Form First

The rule applies *only* to an indeterminate form. Ignore that and it quietly gives the wrong answer:

$$\lim_{x\to 0} \frac{x+1}{2x+1} = \frac{1}{1} = 1$$

But differentiating top and bottom anyway gives

$$\lim_{x\to 0} \frac{1}{2} = \frac{1}{2}$$

which is simply not the limit ✗

So: check the form before every application - including the second and third one, since a limit can leave the indeterminate case after a single step.

Forms You Rewrite First

Two more forms are indeterminate but not quotients. Both become quotients with a little algebra.

$0 \cdot \infty$ - divide by the reciprocal:

$$\lim_{x \to 0^+} x \ln x = \lim_{x \to 0^+} \frac{\ln x}{1/x} \overset{\text{L'H}}{=} \lim_{x \to 0^+} \frac{1/x}{-1/x^2} = \lim_{x \to 0^+} (-x) = 0$$

1^∞ - take logs, apply the rule, then exponentiate back.

Where This Shows Up

$$\lim_{n\to\infty}\left(1+\frac{x}{n}\right)^n$$

is of the form 1^∞. Take logs:

$$\ln L = \lim_{n\to\infty} n\ln\left(1+\frac{x}{n}\right) = \lim_{n\to\infty}\frac{\ln(1+x/n)}{1/n} \overset{\text{L'H}}{=} \lim_{n\to\infty}\frac{\frac{-x/n^2}{1+x/n}}{-1/n^2} = x$$

So $L = e^x$ - a limit you have almost certainly seen as the definition of e (set $x = 1$), and one that does real work two slides from now.

From Binomial to Poisson

A Poisson distribution counts events that are individually rare but have many chances to happen - emails in an hour, crimes in one sector of a city:

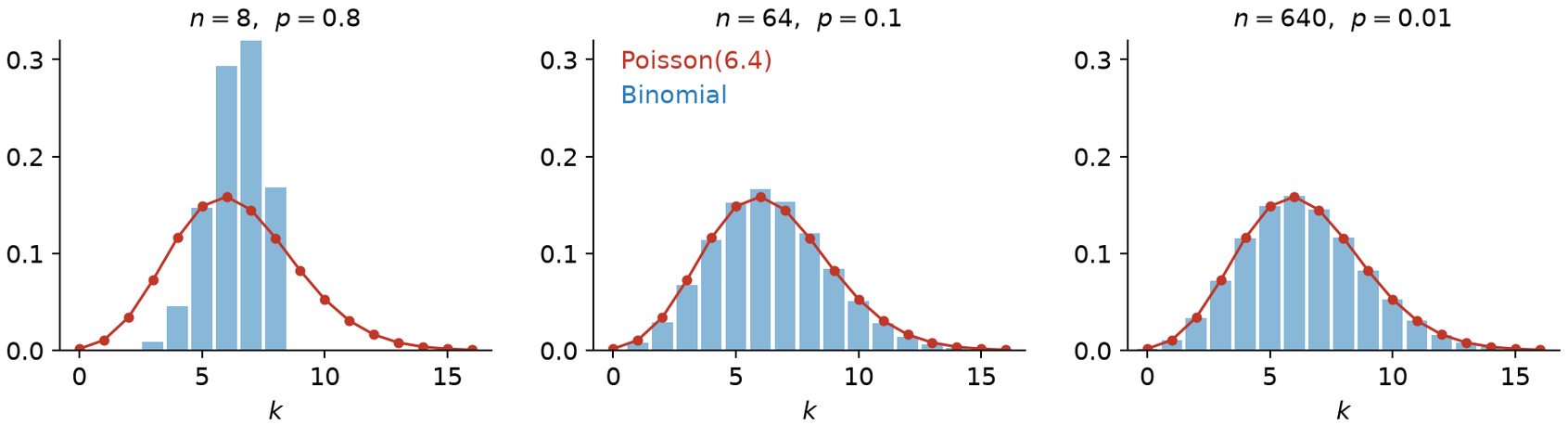
$$P(X = k) = \frac{e^{-\lambda}\,\lambda^k}{k!} \qquad \lambda = \text{the expected count}$$

Where does that $e^{-\lambda}$ come from? Take a $\mathrm{Binomial}(n, p)$, hold $\lambda = np$ fixed, and let $n \to \infty$:

$$\binom{n}{k}\left(\frac{\lambda}{n}\right)^k\left(1-\frac{\lambda}{n}\right)^{n-k} = \underbrace{\frac{n(n-1)\cdots(n-k+1)}{n^k}}_{\to\ 1} \cdot \frac{\lambda^k}{k!} \cdot \underbrace{\left(1-\frac{\lambda}{n}\right)^{n}}_{\to\ e^{-\lambda}} \cdot \underbrace{\left(1-\frac{\lambda}{n}\right)^{-k}}_{\to\ 1}$$

The outer two factors are bookkeeping - a **fixed** number of terms, each tending to 1. The middle one is the limit we just did, at $x = -\lambda$, and it is the only place the e can come from.

From Binomial to Poisson

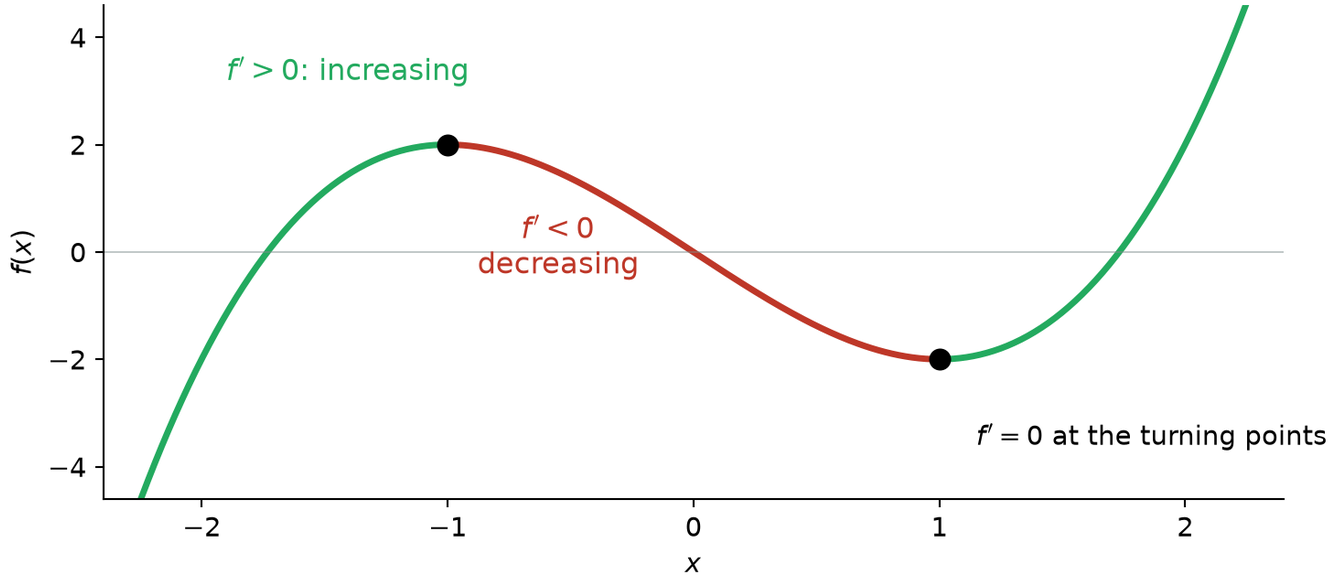


Same expected count $\lambda = 6.4$ in all three: more trials, each of them rarer. With $n = 8$ the binomial cannot even reach $k > 8$; by $n = 640$ the two are hard to tell apart.

Part IV: Derivatives and the Shape of a Graph

What the First Derivative Tells You

The first derivative tells you whether the function is increasing, decreasing, or neither at a point.

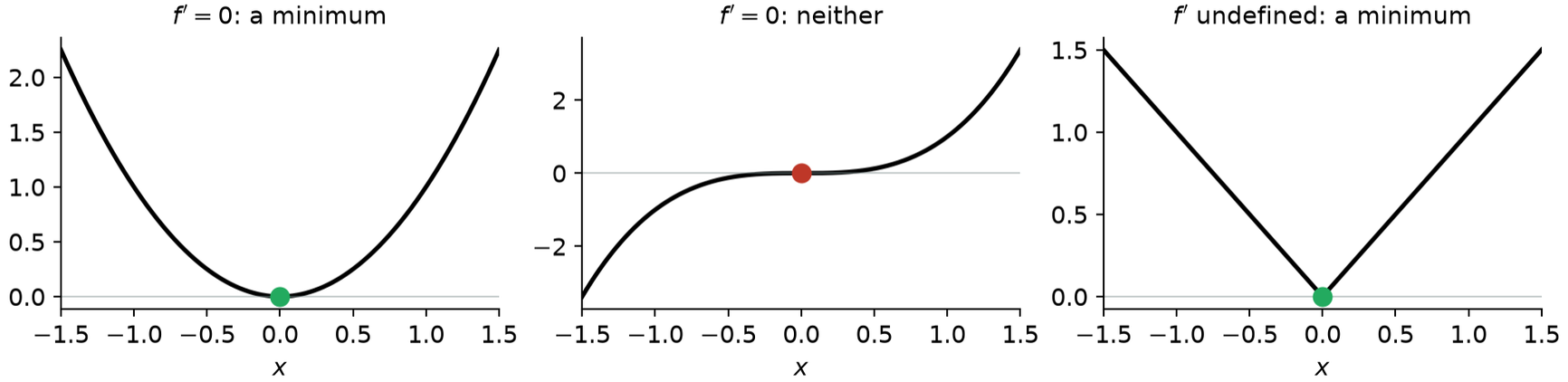


Critical Points

A point $x = c$ in the domain of f is a critical point if either

$$f'(c) = 0 \qquad \text{or} \qquad f'(c) \text{ does not exist}$$

Every local maximum and minimum occurs at a critical point - but **not** every critical point is a local max or min:



The First Derivative Test

Let c be a critical point of a continuous function f. Look at the **sign of f' on either side of c**:

Sign of f' left of c	**Sign of f' right of c**	**Then f has at c**
$+$ (increasing)	$-$ (decreasing)	a local **maximum**
$-$ (decreasing)	$+$ (increasing)	a local **minimum**
$+$	$+$	neither
$-$	$-$	neither

In words: the function has a peak where it stops climbing and starts falling, and a trough where it stops falling and starts climbing.

Worked Example

Find the critical points of $f(x) = x^3 - 3x^2 - 9x + 1$ and classify them.

Step 1 - differentiate and set to zero:

$$f'(x) = 3x^2 - 6x - 9 = 3(x-3)(x+1) = 0 \quad \Longrightarrow \quad x = -1,\ x = 3$$

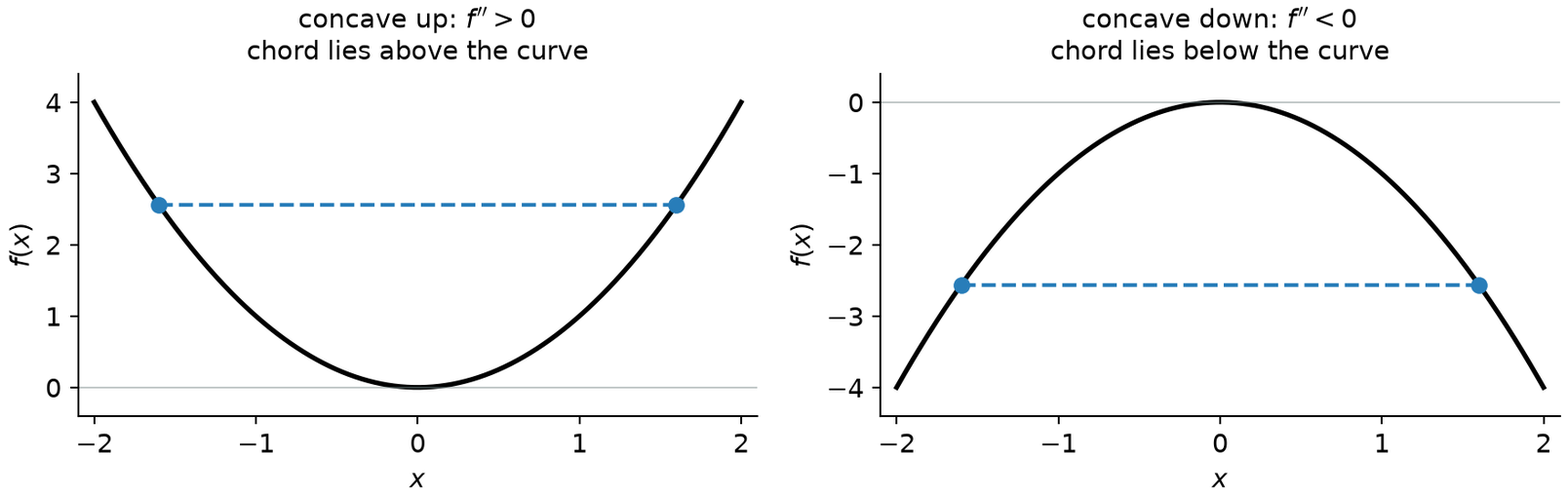
Step 2 - check the sign of f' in the three resulting intervals:

Interval	Test point	f'	f is
$(-\infty, -1)$	$x = -2$	$3(-5)(-1) = +15$	increasing
$(-1, 3)$	$x = 0$	$3(-3)(1) = -9$	decreasing
$(3, \infty)$	$x = 4$	$3(1)(5) = +15$	increasing

So $x = -1$ is a local **maximum** ($+ \to -$) and $x = 3$ a local **minimum** ($- \to +$).

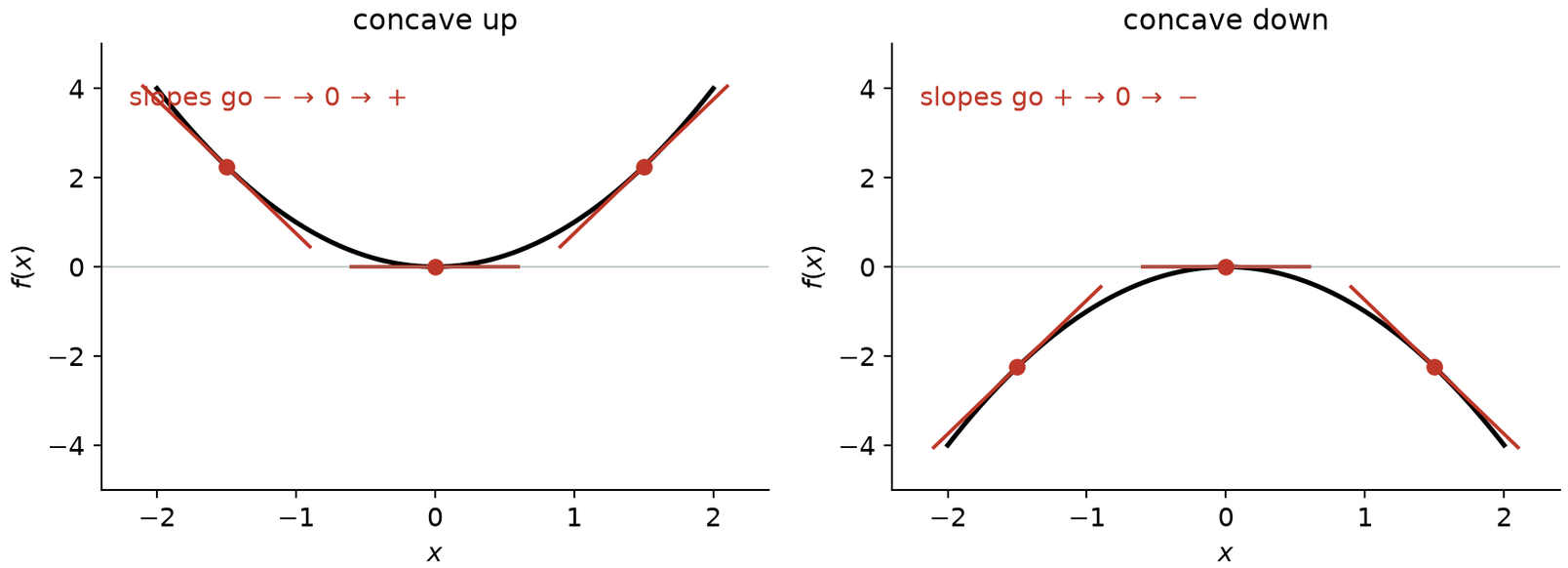
What the Second Derivative Tells You

The second derivative - the rate of change *of the rate of change* - tells us whether the function is concave up or concave down.



Concavity and the Slope

Another way to read it: concave up means the *slope itself* is increasing.



A point where concavity flips is an inflection point: there $f''(x) = 0$.

The Second Derivative Test

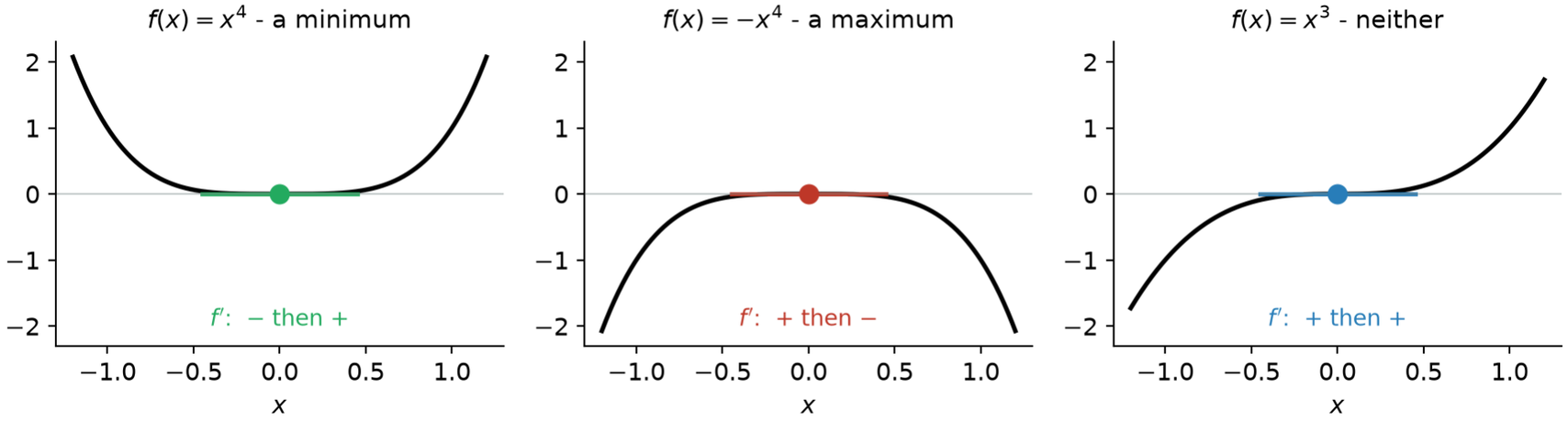
Suppose f'' is continuous and $f'(c) = 0$. Then:

Condition	Conclusion
$f''(c) > 0$	f has a local **minimum** at c (a valley)
$f''(c) < 0$	f has a local **maximum** at c (a peak)
$f''(c) = 0$	the test is **inconclusive** - fall back on the first derivative test

It is usually faster than the first derivative test: one evaluation instead of a sign analysis on every interval. The price is the inconclusive case.

What “Inconclusive” Looks Like

All three of these have $f'(0) = 0$ **and** $f''(0) = 0$ - yet one is a minimum, one a maximum, and one neither:



So $f''(c) = 0$ tells us nothing: the curvature vanishes at that one point, and a single point cannot say which way the curve leaves it. The first derivative test reads a whole neighbourhood instead - the sign of f' either side - so it always decides.

Back to the Worked Example

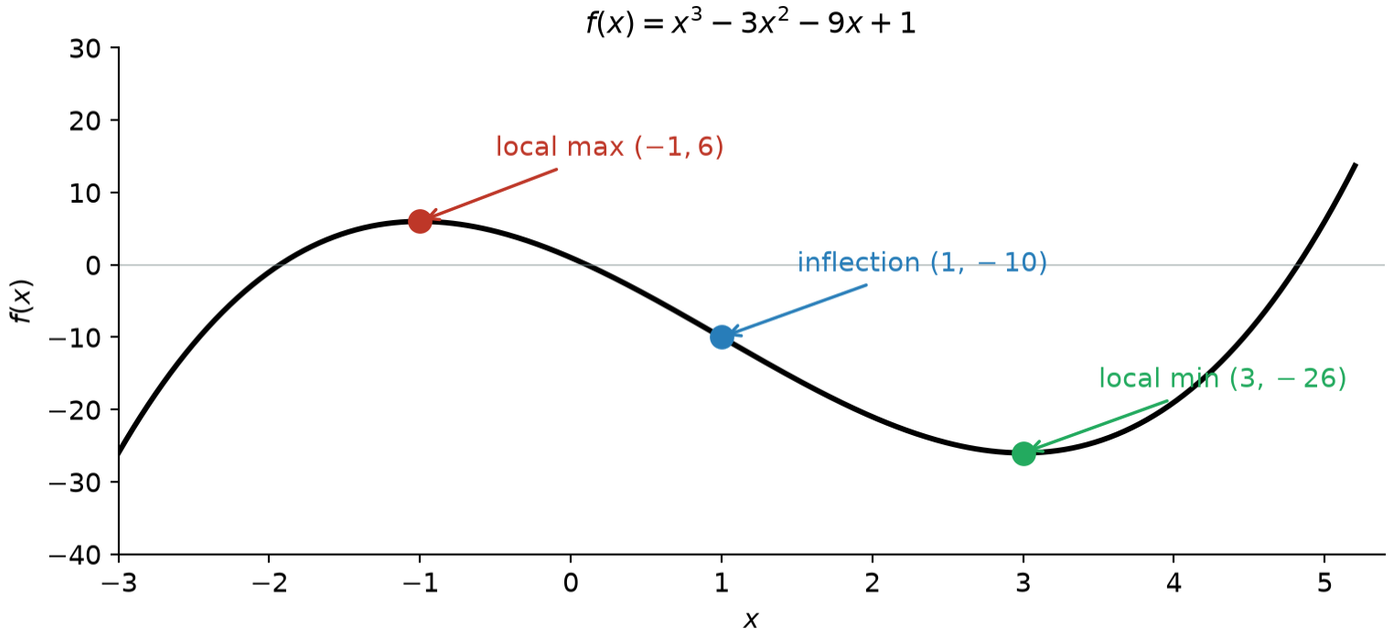
For $f(x) = x^3 - 3x^2 - 9x + 1$ we had $f'(x) = 3x^2 - 6x - 9$, so

$$f''(x) = 6x - 6$$

- $f''(-1) = -12 < 0 \;\Rightarrow$ local **maximum** at $x = -1$, where $f(-1) = 6$
- $f''(3) = 12 > 0 \;\Rightarrow$ local **minimum** at $x = 3$, where $f(3) = -26$
- $f''(x) = 0$ at $x = 1$: an **inflection point**, where $f(1) = -10$ and $f'(1) = -12$

Same conclusions as the first derivative test - plus the inflection point for free.

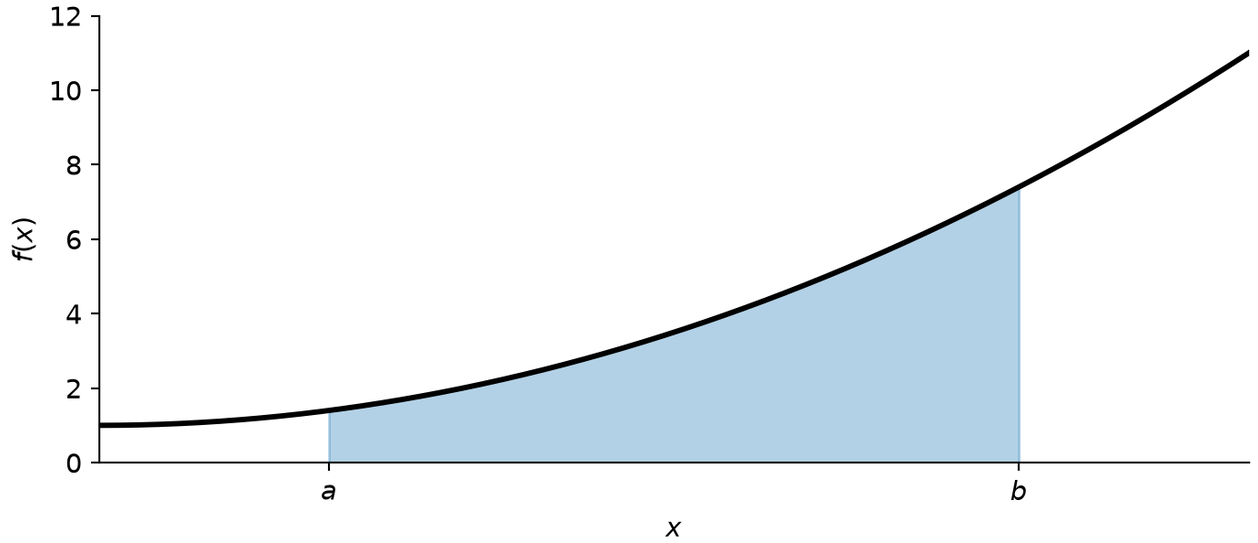
Back to the Worked Example



Part V: Integration

The Definite Integral

We are interested in the area under this curve over the shaded region:

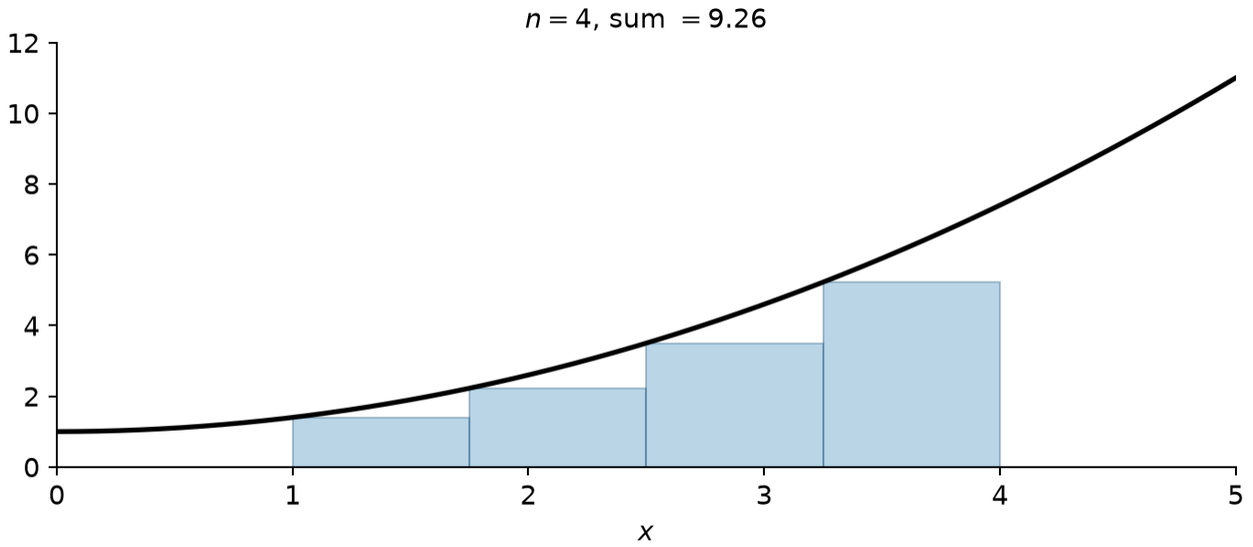


which we write

$$A = \int_a^b f(x)\,dx$$

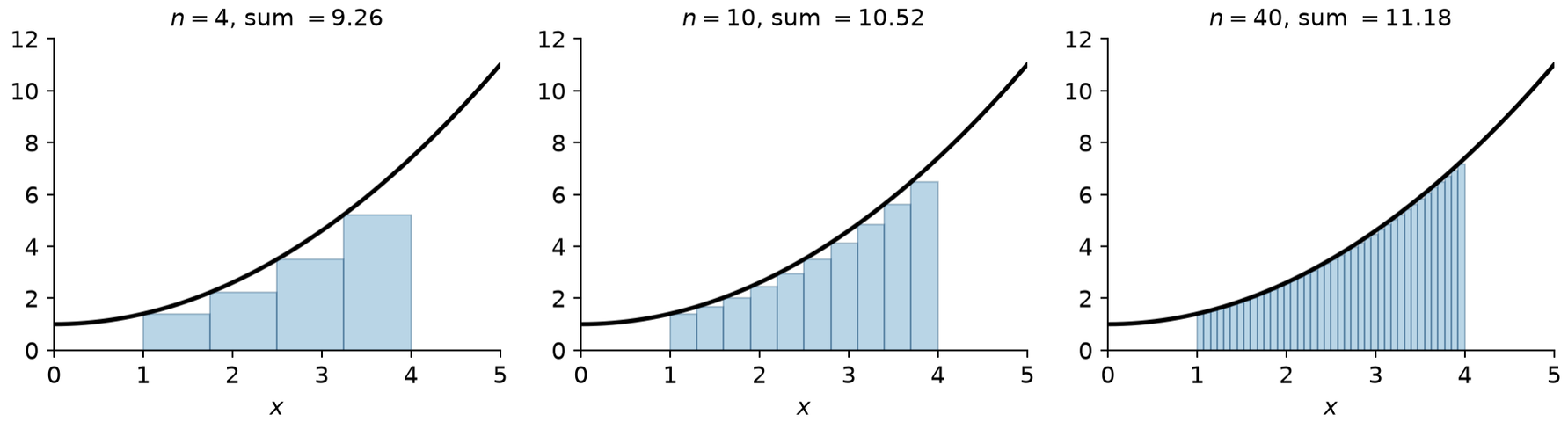
Left-Endpoint Approximation

This area can be approximated with rectangles whose heights are read off the left endpoint of each subinterval:



Each rectangle has area $f(x_i^*)\,\Delta x$ - a height times a width.

More Rectangles, Better Approximation



The exact area is $\int_1^4 (0.4x^2 + 1)\,dx = 11.4$ - the sums climb towards it as the rectangles get thinner.

The Riemann Sum

The approximation is called a Riemann sum:

$$\sum_{i=1}^{n} f(x_i^*)\,\Delta x$$

where Δx is the width of each subinterval $[x_{i-1}, x_i]$ and x_i^* is any point in that subinterval.

Then the area under the curve is

$$A = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\,\Delta x = \int_a^b f(x)\,dx$$

This is the definite integral.

The Fundamental Theorem of Calculus, Part 1

Part 1 establishes the relationship between the derivative and the integral.

> If f is continuous on $[a, b]$ and F is defined by
>
> $$F(x) = \int_a^x f(t)\,dt$$
>
> then $F'(x) = f(x)$ on $[a, b]$.

In words: **integration and differentiation undo each other.** F is an antiderivative of f.

This is exactly the relationship between a CDF and a PDF, which is where we pick it up in Lecture 5:

$$F(x) = \int_{-\infty}^x f(t)\,dt \qquad \Longleftrightarrow \qquad f(x) = F'(x)$$

The Fundamental Theorem of Calculus, Part 2

Part 2 tells us how to actually *evaluate* a definite integral.

> If f is continuous on $[a, b]$ and F is any antiderivative of f, then
>
> $$\int_a^b f(x)\,dx = F(b) - F(a)$$

So the whole job reduces to: **find an antiderivative, then subtract its values at the two endpoints.** No limits of sums required.

Example: Computing a Definite Integral

$$\int_0^3 (x^2 - 1)\,dx$$

Step 1 - apply Part 1: find an antiderivative. Reversing the power rule, $\frac{d}{dx}\frac{x^{n+1}}{n+1} = x^n$, so

$$F(x) = \frac{x^3}{3} - x$$

Step 2 - apply Part 2: evaluate at the endpoints and subtract.

$$\int_0^3 (x^2 - 1)\,dx = F(3) - F(0) = \left(\frac{27}{3} - 3\right) - (0 - 0) = 9 - 3 = 6$$

Beyond Polynomials

For everything that is not a polynomial there are many integration techniques, and the right choice depends on the application:

1. **Find the antiderivative** in terms of other, simpler functions
    - substitution, integration by parts, partial fractions, trigonometric substitution

2. **Numerical integration** - approximating a definite integral when no elementary antiderivative exists
    - midpoint rule, trapezoidal rule, Simpson's rule
    - Gaussian quadrature, Bayesian quadrature
    - Monte Carlo integration
    - sparse grid integration

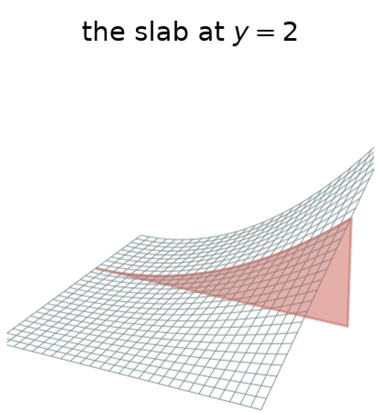
The Normal PDF is the standard example of case 2: $\int e^{-x^2/2}\,dx$ has no elementary antiderivative, which is why Φ comes out of a table or a library call.

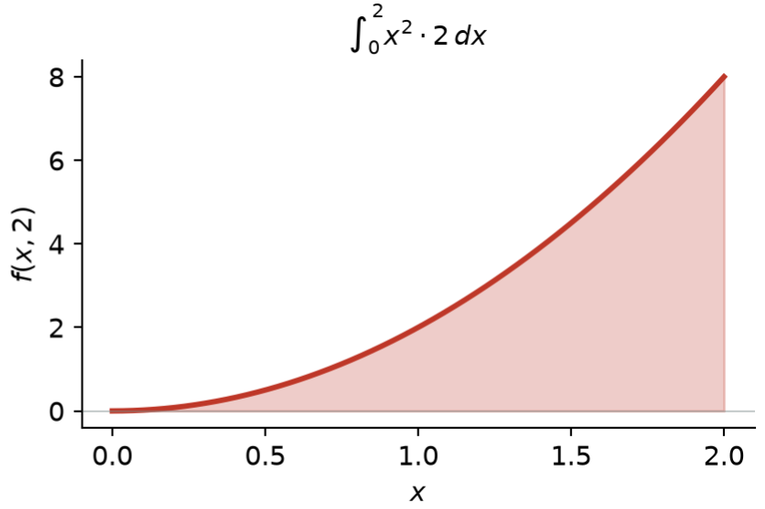
Integrating Over Two Variables

A definite integral measures the area under a curve. Over two variables it measures the volume under a surface:

$$\int_c^d \int_a^b f(x,y)\,dx\,dy$$

And it is computed by the same move that defined the partial derivative - **one variable at a time, holding the other fixed**:





The inner integral fixes y and sweeps x: the area of one slab. The outer one then adds the slabs as y moves.

Integrating Over Two Variables: Example

$$\int_0^3 \int_0^2 x^2 y \, dx \, dy$$

Inner integral - y is a constant, so pull it out and integrate x^2 as usual:

$$\int_0^2 x^2 y \, dx = y \left[\frac{x^3}{3} \right]_0^2 = \frac{8}{3} y$$

Outer integral - now an ordinary one-variable integral in y:

$$\int_0^3 \frac{8}{3} y \, dy = \frac{8}{3} \left[\frac{y^2}{2} \right]_0^3 = \frac{8}{3} \cdot \frac{9}{2} = 12$$

Integrating in the other order gives 12 as well - over a rectangle, with a well-behaved f, the order is yours to choose.

Why We Need This

Next lecture, a pair of continuous random variables has a joint PDF $f_{X,Y}(x, y)$, and everything we did with one variable is done with two:

- A probability is a volume: $P\big((X, Y) \in A\big) = \iint_A f_{X,Y}(x, y)\, dx\, dy$
- A marginal density integrates the other variable away: $f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y)\, dy$
- And the joint PDF is recovered from the joint CDF by the mixed second partial: $f_{X,Y} = \dfrac{\partial^2 F_{X,Y}}{\partial x\, \partial y}$

Sum over one index, integrate over one variable - it is the discrete marginalisation of Lecture 3 with an integral sign.

Summary - calculus

Concept	Key formula
Derivative	$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$
Power rule	$\frac{d}{dx} x^n = n x^{n-1}$
Product rule	$(fg)' = f'g + fg'$
Quotient rule	$\left(\frac{f}{g}\right)' = \frac{gf' - fg'}{g^2}$
Chain rule	$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$
Exponential / log	$\frac{d}{dx} e^x = e^x$; $\frac{d}{dx} \ln x = \frac{1}{x}$
L'Hôpital's rule	$\lim \frac{f}{g} = \lim \frac{f'}{g'}$, for the forms $\frac{0}{0}$ and $\frac{\infty}{\infty}$
Partial derivative	$\frac{\partial f}{\partial x}$: differentiate in x, hold y constant; order of mixed partials is free

Summary - calculus cont.

Concept	Key formula
Critical point	$f'(c) = 0$ or $f'(c)$ undefined
First derivative test	sign change $+ \to -$: max; $- \to +$: min
Second derivative test	$f''(c) > 0$: min; $f''(c) < 0$: max; $f''(c) = 0$: inconclusive
Riemann sum	$A = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\Delta x = \int_a^b f(x)dx$
FTC Part 1	$F(x) = \int_a^x f(t)dt \Rightarrow F'(x) = f(x)$
FTC Part 2	$\int_a^b f(x)dx = F(b) - F(a)$
Iterated integral	$\int_c^d \int_a^b f(x,y)\,dx\,dy$ - inner integral holds y fixed

Reading

Herman & Strang, *Calculus, Volume 1*

- Ch. 3.3 - differentiation rules; Ch. 3.4 (p. 270–272), Ch. 3.6 (p. 287–289) - the chain rule
- Ch. 3.9 (p. 323–327) - derivatives of exponentials and logarithms
- Ch. 4.3 (p. 366–375) - maxima and minima; Ch. 4.5 - derivatives and the shape of a graph
- Ch. 4.8 - l'Hôpital's rule and indeterminate forms

Herman & Strang, *Calculus, Volume 2*

- Ch. 1.1 (p. 6–16) - approximating areas
- Ch. 1.2 (p. 24–26, 32–33) - the definite integral and its properties
- Ch. 1.3 (p. 43–48) - the Fundamental Theorem of Calculus, Parts 1 and 2